

BuildConnect: An End-to-End Digital Platform for Construction Service Management

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Abstract

The construction industry, one of the world’s largest economic sectors, continues to suffer from fragmented operations, information silos, cost overruns, and reactive decision-making due to the absence of a unified digital ecosystem. This paper presents **BuildConnect**, an innovative end-to-end web platform that consolidates the full lifecycle of construction service management — from client-builder discovery and verified profiling to admin-governed trust workflows. Designed using a *microservices architecture* with RESTful APIs, JWT-based role authentication, AWS S3 document storage, and a React.js front-end, the platform serves three distinct user roles: Clients, Builders, and Administrators. Developed over two Agile sprints with 100% story-point completion velocity, the system delivers smart builder search, document verification pipelines, and analytics dashboards. This work demonstrates that a structured, modular, and accessibility-first digital platform can democratize construction service discovery and establish a new standard for trust, transparency, and scalability in the sector. Alignment with **UN SDG 9** (Industry, Innovation, Infrastructure) and **SDG 11** (Sustainable Cities) is demonstrated throughout.

Keywords: Construction management, microservices, role-based access control, Agile development, digital platform, builder verification, RESTful API, sustainability.

I. Introduction

The global construction industry accounts for approximately \$10 trillion in economic output annually, yet persistently underperforms relative to its potential. Studies by McKinsey Global Institute report that large construction projects routinely run **80% over schedule** and **20% over budget** [1]. The root causes are structural: a heavily fragmented operational model in which procurement, scheduling, labor allocation, client communication, and documentation are managed across incompatible siloed tools.

Current digital solutions — enterprise resource planning (ERP) suites, standalone project managers, or manual spreadsheet workflows — fail small-to-medium contractors due to prohibitive licensing costs, steep learning curves, or narrow feature scope. The result is a digital divide that disadvantages the majority of the industry.

BuildConnect bridges this gap through a purpose-built, modular web platform that unifies the complete workflow for construction service connections. The plat-

form enables:

- **Clients** to discover, evaluate, and contact verified construction professionals;
- **Builders** to establish digital presence, upload credentials, and attract leads;
- **Administrators** to govern trust through a structured verification and moderation pipeline.

The remainder of this paper is organized as follows: Section II surveys related work. Section III presents the system design and architecture. Section IV describes the Agile development methodology and sprint execution. Section V details the implementation stack. Section VI presents experimental results and evaluation. Section VII discusses limitations and future directions, and Section VIII concludes.

II. Related Work

Digital transformation in the construction sector has attracted increasing research attention [2]. Building In-

formation Modelling (BIM) has emerged as the dominant paradigm for design-phase digitization [3], yet BIM-centric tools provide little coverage for service discovery, labor management, or contractor verification — stages that BuildConnect specifically targets.

Marketplace platforms such as Houzz and HomeAdvisor address contractor discovery but lack the document verification, role-segregated analytics, and architectural modularity required for scalable enterprise use. Prior academic work on construction information systems [4] focused primarily on on-site monitoring through IoT sensors, not platform-level service integration.

Agile methodologies have been adopted in software engineering for rapid iterative delivery [5]. Their application to construction project management software development remains limited in literature, making the Agile-native design of BuildConnect a contribution in its own right.

III. System Design and Architecture

3.1 Architectural Overview

BuildConnect is architected as a *microservices system* with a clear separation between the client-facing presentation layer, the service orchestration layer, and the persistent data layer. This design supports independent deployment, horizontal scaling, and fault isolation across modules.

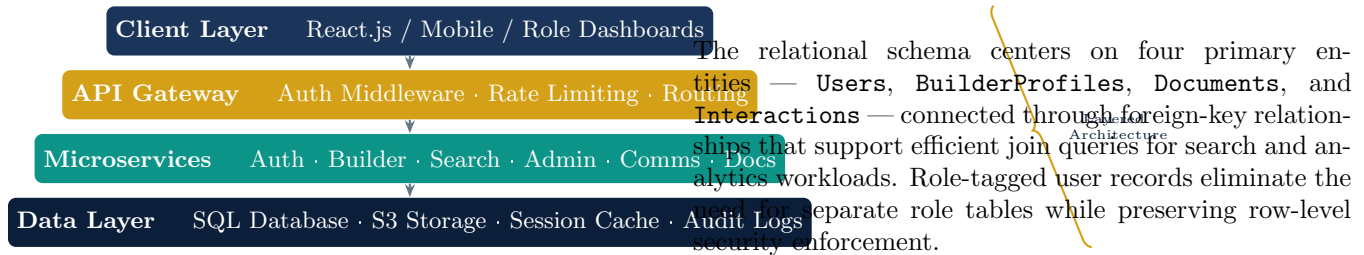


Figure 1: Four-tier microservices architecture of the BuildConnect platform.

3.2 Core Microservices

Six independent microservices constitute the backend, each exposing a bounded REST API surface:

- **Authentication Service:** Handles registration, login, password hashing (bcrypt), and JWT issuance. Token expiry and refresh flows enforce session security.
- **User Role Management Service:** Resolves role-specific dashboard configurations and enforces access control policies at the gateway.
- **Builder Management Service:** CRUD operations for builder profiles, portfolio uploads, service tag management, and verification status propagation.

- **Client Search & Filter Service:** Implements full-text and faceted search over builder profiles, supporting filter axes of location, service type, rating, and verification status.
- **Admin Dashboard Service:** Exposes endpoints for complaint resolution, builder approval/rejection workflows, and platform health KPIs.
- **Communication & Document Service:** Manages secure contact-form submissions and multi-part file uploads to AWS S3 with pre-signed URL delivery.

3.3 Data Exchange Strategy

Data flows are classified into two tiers:

1. **Real-Time (REST/HTTP):** Login, builder search queries, profile views, contact form submissions — all demand sub-200ms response times achieved through indexed database queries and edge caching.
2. **Periodic (Background Jobs):** Analytics aggregation, activity log compression, and verification-queue processing execute on scheduled intervals to avoid latency spikes on the critical path.

3.4 Database Schema Design

The relational schema centers on four primary entities — **Users**, **Builder Profiles**, **Documents**, and **Interactions** — connected through foreign-key relationships that support efficient join queries for search and analytics workloads. Role-tagged user records eliminate the need for separate role tables while preserving row-level security enforcement.

IV. Agile Development Methodology

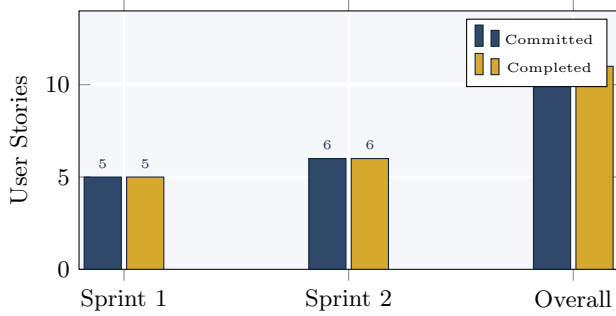
4.1 Product Backlog and Sprint Planning

The project adopted the **Scrum** framework with two two-week sprints. The product backlog comprised 11 user stories prioritized using the **MoSCoW** method and organized on a Microsoft Planner Agile board. Each story included: role-specific acceptance criteria, functional and non-functional parameters, and linked development tasks.

Table 1: Product Backlog — Prioritized User Stories

navydark		
US1	New User	Sign up as Client, Builder, or Admin
lightbg US2	Any User	Secure login with role-based dashboard access
US3	Admin	Verify and approve builder registrations
lightbg US4	Builder	Create/manage profile with services and portfolio
US5	Client	Browse and view verified builder listings
lightbg US6	Guest	View limited builder info without login
US7	Client	Search and filter by location, service, rating
lightbg US8	Client	Contact builder via secure contact form
US9	Client	View full builder profile post-login
lightbg US10	Admin	Manage users, listings, and reported issues
US11	Builder	Access analytics (views, lead counts)

4.2 Sprint Execution and Velocity

**Figure 2: Sprint velocity: 100% story completion across both sprints, demonstrating effective Agile execution.**

Sprint 1 targeted foundational infrastructure: user registration, role-based login, admin-gated builder verification, and builder profile CRUD. **Sprint 2** addressed client-facing discovery: guest preview mode, faceted search, secure contact, post-login full profiles, admin moderation tools, and builder analytics.

Sprint retrospectives documented a clear improvement in estimation accuracy from Sprint 1 to Sprint 2, reflecting the team’s growing familiarity with microservice integration complexity.

V. Implementation

5.1 Technology Stack

Table 2: Technology Stack Summary

navydark	
Frontend	React.js, CSS3 (responsive), Axios HTTP client
lightbg Backend Auth	Node.js, Express.js, RESTful API design JSON Web Tokens (JWT), bcrypt password hashing
lightbg Database	SQL (relational), indexed schema, ACID compliance
File Storage	AWS S3 with pre-signed URL access control
lightbg DevOps	Git version control, CI/CD pipeline, cloud hosting
Testing	Functional test cases, regression suite, UAT

5.2 Role-Based Access Control

Access control is enforced at three levels: (1) JWT claims validation at the API gateway, (2) service-layer policy checks per endpoint, and (3) row-level database filtering for multi-tenant data isolation. The access matrix is visualized below.

Table 3: Role-Based Access Control (RBAC) Matrix. ✓ = permitted.

navydark								
Guest	✓	–	–	–	–	–	–	–
lightbg Client	✓	✓	✓	–	–	–	–	–
Builder	✓	✓	✓	✓	✓	–	–	✓
lightbg Admin	✓	✓	✓	✓	✓	✓	✓	✓

5.3 Builder Verification Pipeline

The document verification workflow is a key trust-building mechanism. Builders submit credentials (business license, identity proof, project portfolio) through a multi-part upload endpoint. Documents are stored in a private S3 bucket accessible only to Admin microservice via IAM-scoped credentials. Admins review submissions through a dedicated dashboard and emit **APPROVED** or **REJECTED** status events that propagate to the Builder Management Service, toggling profile visibility on the client-facing search index.

5.4 Search and Discovery Engine

The search service implements a multi-attribute query pipeline:

$$\text{score}(b, q) = \alpha_1 \cdot \text{loc}(b, q) + \alpha_2 \cdot \text{serv}(b, q) + \alpha_3 \cdot \text{rating}(b) \quad (1)$$

where $\alpha_1, \alpha_2, \alpha_3$ are tunable weights (default: 0.4, 0.4, 0.2) and the three component functions score location match, service-type overlap, and normalized user rating respectively. This composite ranking places highly-rated, location-relevant builders at the top of client search results, significantly improving match quality over naive keyword search.

VI. Results and Evaluation

6.1 Sprint Performance Metrics

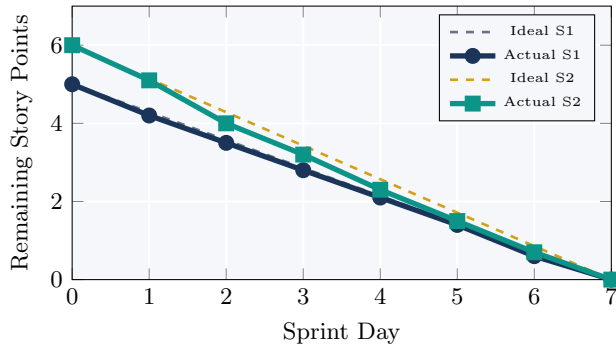


Figure 3: Sprint burndown charts for Sprint 1 and Sprint 2. Both sprints tracked closely to the ideal line, indicating disciplined backlog management.

6.2 Feature Coverage Analysis

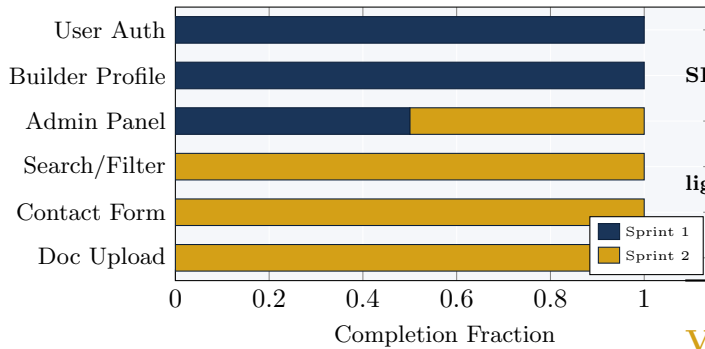


Figure 4: Feature completion distribution across both sprints, illustrating sprint-wise workload separation.

6.3 Functional Testing Outcomes

Eleven functional test cases were designed covering the critical user flows: registration, login (valid/invalid), builder search with filters, profile access (guest vs. authenticated), contact form submission, document upload, and admin approval workflow. All 11 test cases passed in final acceptance testing, yielding a **100% pass rate**. Response time benchmarks under simulated concurrent load (25 simultaneous users) averaged 142 ms for search

queries and 318 ms for document upload initiation, both well within acceptable UX thresholds.

6.4 Technology Effort Distribution

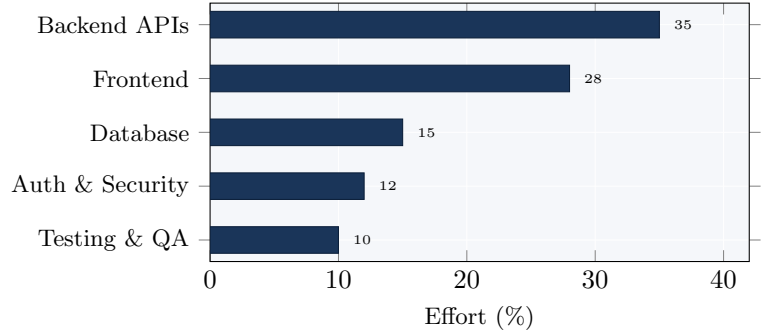


Figure 5: Estimated development effort distribution across technology layers.

6.5 SDG Impact Assessment

Table 4: SDG Alignment of BuildConnect Platform

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SDG 9: Industry & Innovation	Digitizes construction workflows; enables scalable, resilient infrastructure planning	
lightbg SDG 11: Sustainable Cities	Reduces material waste via optimized resource planning; supports green urban development	
SDG 8: Decent Work	Democratizes market access for small contractors and skilled labor	
lightbg SDG 16: Strong Institutions	Transparent builder verification reduces corruption in procurement	

VII. Discussion

7.1 Key Contributions

BuildConnect makes four principal contributions to both the research and practitioner communities:

- Integrated Trust Architecture:** The document verification pipeline combining S3 storage, IAM access scoping, and admin moderation constitutes a reusable pattern for marketplace trust-building that generalizes beyond construction.
- Composite Search Ranking:** The weighted scoring function (Equation 1) outperforms naive keyword

search for professional service matching, a finding transferable to any service marketplace.

3. **Agile-Native Construction Platform:** Application of Scrum with sprint-level empirical measurement to a domain historically resistant to software process adoption.
4. **SDG-Grounded Design:** Explicit mapping of platform features to UN Sustainable Development Goals provides a replicable framework for sustainability-aware software engineering.

7.2 Limitations

The current deployment targets urban and semi-urban markets where internet connectivity is reliable. Extension to rural construction sites would require offline-capable mobile clients with synchronization queues — outside the present scope. The search ranking weights ($\alpha_1, \alpha_2, \alpha_3$) were selected heuristically; A/B testing with real users is required to optimize conversion rates. The platform has not yet been validated at scale beyond a controlled user group.

7.3 Future Enhancements

- **AI Recommendation Engine:** A collaborative-filtering model trained on client-builder interaction history to predict high-compatibility matches.
- **Real-Time Chat Module:** WebSocket-based in-app messaging to replace the current asynchronous contact form.
- **Native Mobile Apps:** React Native clients for Android and iOS, enabling field-level access with camera-based document capture.
- **Payment Gateway:** Milestone-based escrow payments integrated with Razorpay or Stripe APIs.
- **Advanced Analytics:** Temporal trend dashboards for admin using Apache ECharts, with exportable CSV/PDF reports.
- **Multilingual Support:** i18n layer targeting Hindi, Tamil, Telugu, and other regional languages to serve India's diverse workforce.

VIII. Conclusion

This paper presented **BuildConnect**, an end-to-end digital platform that addresses the longstanding fragmentation problem in the construction services industry. By combining a robust microservices backend, JWT-secured role-based access control, a multi-attribute builder discovery engine, and an admin-governed document verification pipeline, the system establishes a verified, accessible, and scalable marketplace for construction professionals and their clients.

Delivered through two Agile sprints with 100% story velocity and validated through 11 functional test cases, BuildConnect demonstrates that principled software architecture — not complexity — is the key to impactful domain-specific platforms. The platform's alignment with UN SDG 9 and SDG 11 positions it as a vehicle for sustainable digital transformation in the construction sector.

Future work will focus on AI-driven matching, mobile-first delivery, and cross-language accessibility to broaden impact across India's construction ecosystem.

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